

ML With BigData

(Project Proposal)

Scalable Real-Time Air Quality Prediction and Pollution Alert System for Indian Cities

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ML With BigData	1
Scalable Real-Time Air Quality Prediction and Pollution Alert System for Indian Cities	1
1. Broad Overview	3
2. Objectives	3
3. Tech Stacks to be Used and Justification	3
4. Deliverables	3
5. Details of Data Source, Format, and Size	4
Asit Jain(M25DE1049)	5
Avinash Singh(M25DE1024)	5
Prashant Kumar Mishra(M25DE1063)	5

1. Broad Overview

Air pollution is a major public health concern in India, especially in cities like Delhi, Mumbai, and Bengaluru. Government monitoring stations continuously generate high-frequency environmental data (PM2.5, PM10, NO₂, CO, etc.). This project aims to build a **Big Data-driven machine learning system** that ingests large-scale air quality data from Indian cities, performs distributed processing, and predicts short-term AQI (Air Quality Index) levels. The system will also simulate real-time alerts for hazardous pollution conditions using streaming frameworks. The project demonstrates how scalable ML and Big Data technologies can address real societal challenges in India.

2. Objectives

- To build a distributed data ingestion and preprocessing pipeline for large-scale Indian air quality datasets.
- To develop ML models for short-term AQI prediction (1–24 hour forecasting).
- To implement a streaming-based pollution alert system for real-time risk detection.

3. Tech Stacks to be Used and Justification

The project will use **Apache Spark** for distributed preprocessing and model training due to its scalability and Spark MLlib integration. **Hadoop (HDFS)** will store raw and processed environmental datasets. **Apache Kafka** will simulate real-time sensor data ingestion. Python (PySpark) will be used for ML model implementation (Regression, Random Forest, Gradient Boosting). Visualization dashboards may be built using Streamlit or Power BI. This stack ensures fault tolerance, scalability, and suitability for handling multi-year environmental datasets.

4. Deliverables

- End-to-end Big Data architecture diagram
- Distributed ETL pipeline implementation
- Feature engineering and time-series analysis report
- Implementation of at least two forecasting models
- Model evaluation metrics (RMSE, MAE, R^2)
- Real-time pollution alert simulation using streaming
- Final technical report (10- 15 pages)
- Documented GitHub repository

5. Details of Data Source, Format, and Size

The dataset will be sourced from:

- **Source:** IQAir API For Real Time and Kaggle
- **Format:** CSV / API-based data
- **Data Size:** Multi-year data across 300+ monitoring stations
- **Features:** PM2.5, PM10, NO₂, SO₂, CO, Temperature, Humidity, Timestamp, Location

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- Data ingestion (API pulling + Kafka streaming simulation)
- HDFS storage setup and data scaling
- Streaming alert pipeline implementation

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- Data cleaning, handling missing values
- Feature engineering (lag features, rolling averages, seasonal components)
- Model training and hyperparameter tuning

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- Distributed ML model implementation using Spark MLlib
- Model evaluation and benchmarking
- Architecture documentation and final report preparation